Knowledge Graphs × LLMs

Tuning LLMs via KG Reasoning

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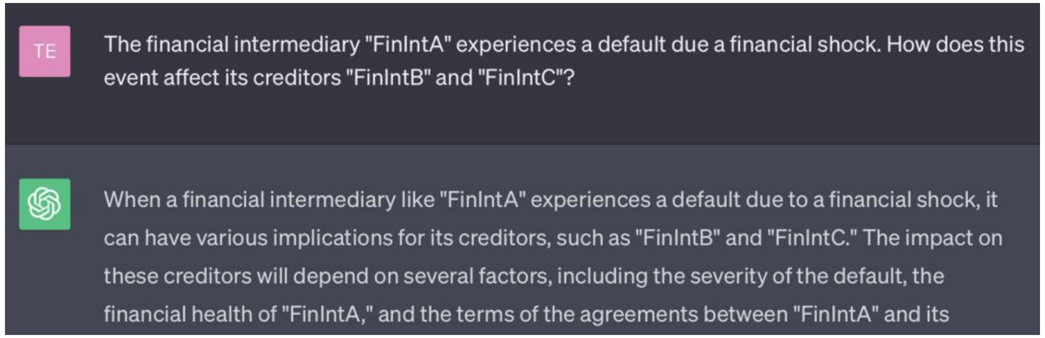
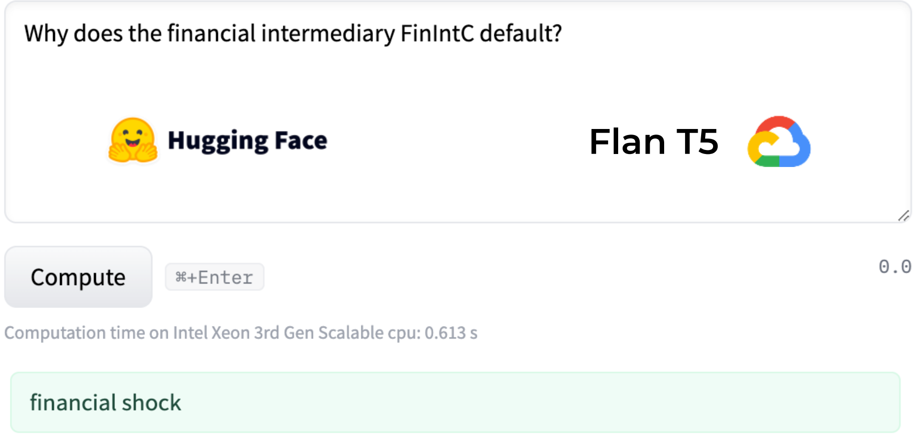
Task-specific fine-tuning

- model customization over **domain-generic** textual corpora to perform **specific NLP tasks**
- good **generalization** capabilities and improved human-style interactions
- lack of **enterprise knowledge** leading to trivially fail **domain-specific** NLP tasks

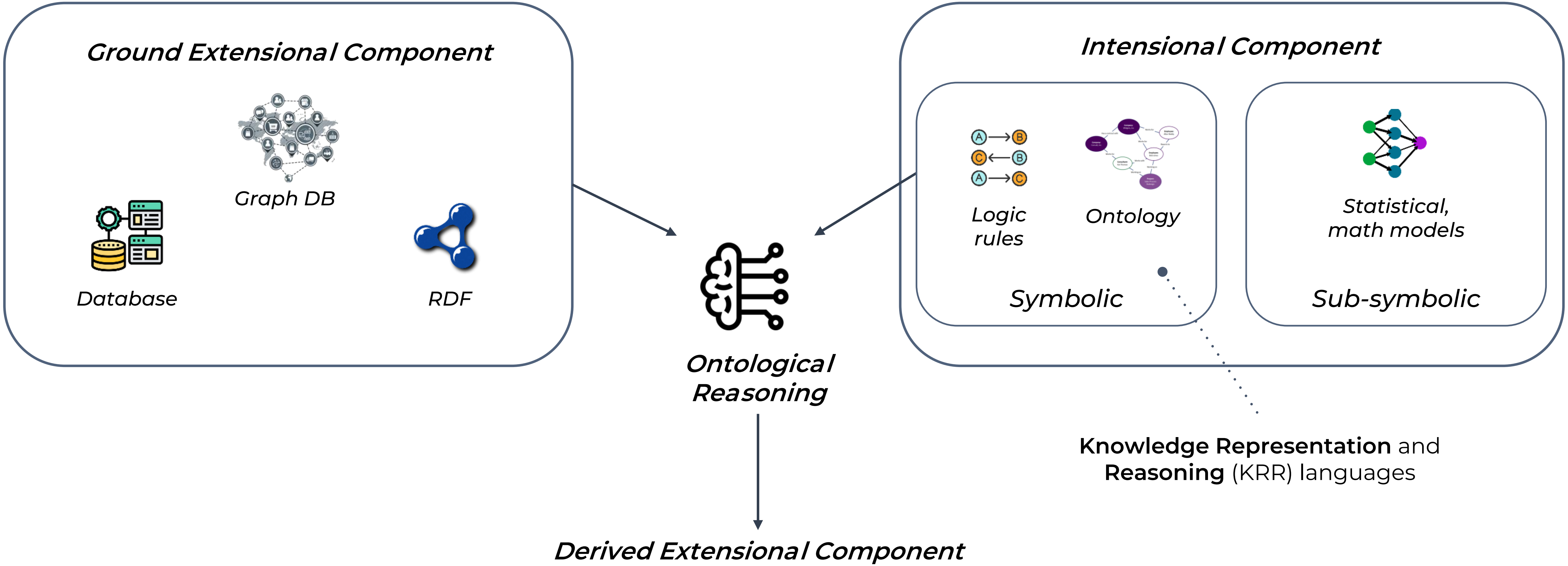


Domain-specific fine-tuning

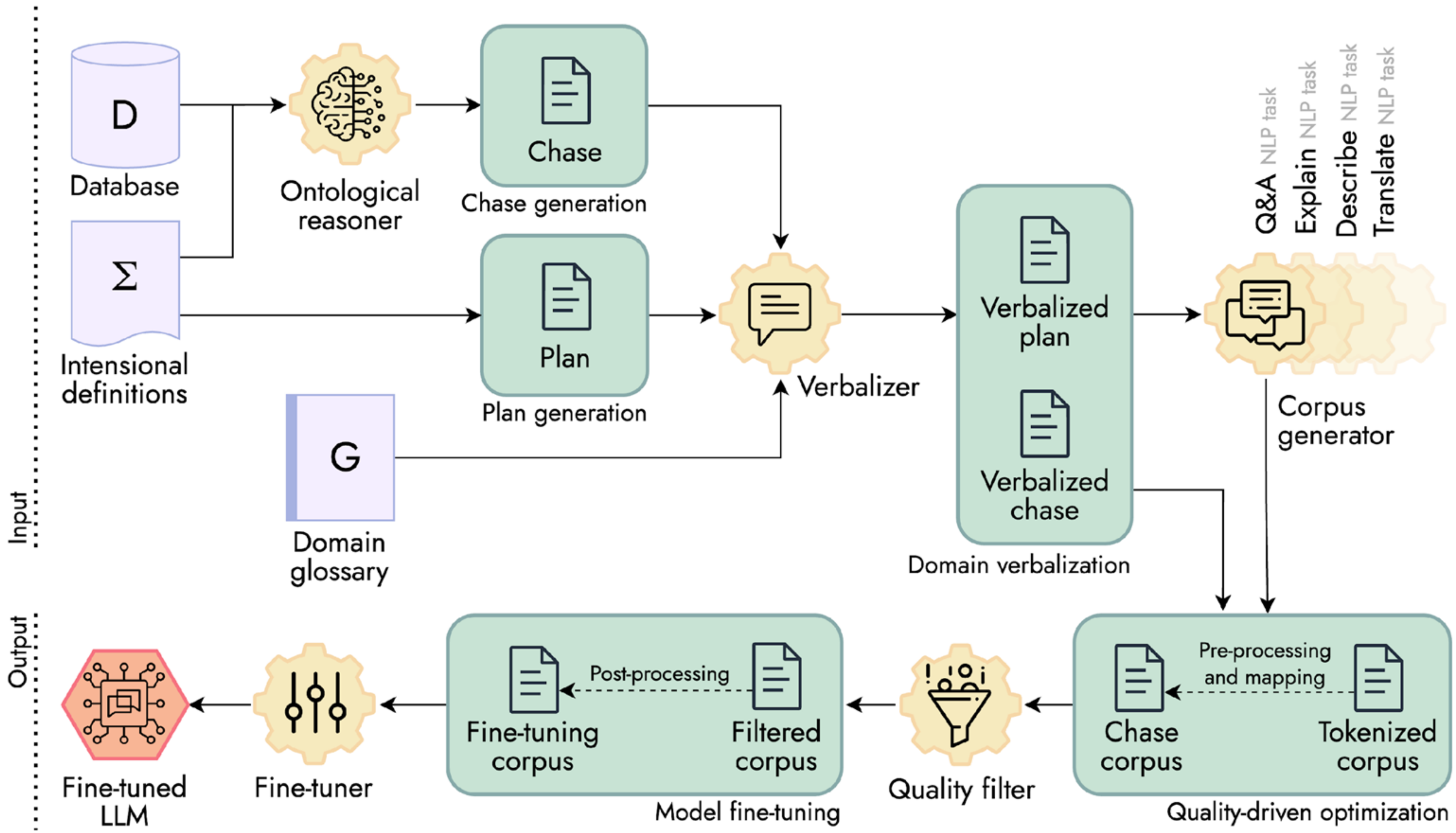
- model customization over **domain data** from enterprise databases for **specific NLP tasks**
- improvements in accuracy but limited to **factual information** of the input databases
- absence of actual **reasoning capabilities**



Liu, Zhuang, et al. (2021). Finbert: A pre-trained financial language representation model for financial text mining. IJCAI
Moiseev, Fedor, et al. (2022). SKILL: structured knowledge infusion for large language models. ACL.
Wu, Shijie, et al. (2023). BloombergGPT: A large language model for finance. CoRR.



EKGs are a cross-cutting topic involving aspects of **data management** (graph databases, file formats, …), **data integration**, **knowledge organization** (constraints, ontologies), and **advanced analytics** (expressive query languages)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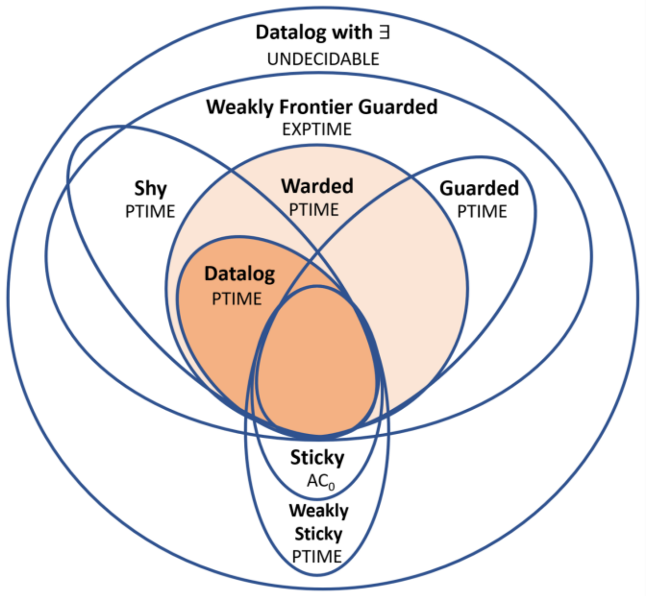
Language syntax

$$\forall x \forall y (\varphi(x,y)) \rightarrow \exists z\ \psi(x,z))$$ *tuple-generating dependencies (**TGDs**)*

*where φ and ψ are conjunctions of atoms that form the **body** and the **head** of the rule, respectively*

Main language features

- **existential** quantification
- monotonic **aggregations**
- stratified **negation**
- **arithmetic** operators
- decidability and **PTIME** in reasoning



Sample program

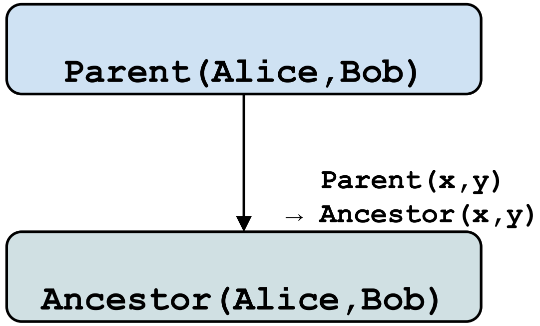
```
Parent(x,y) → Ancestor(x,y) (1)
Parent(x,y), Ancestor(y,z) → Ancestor(x,z) (2)
```

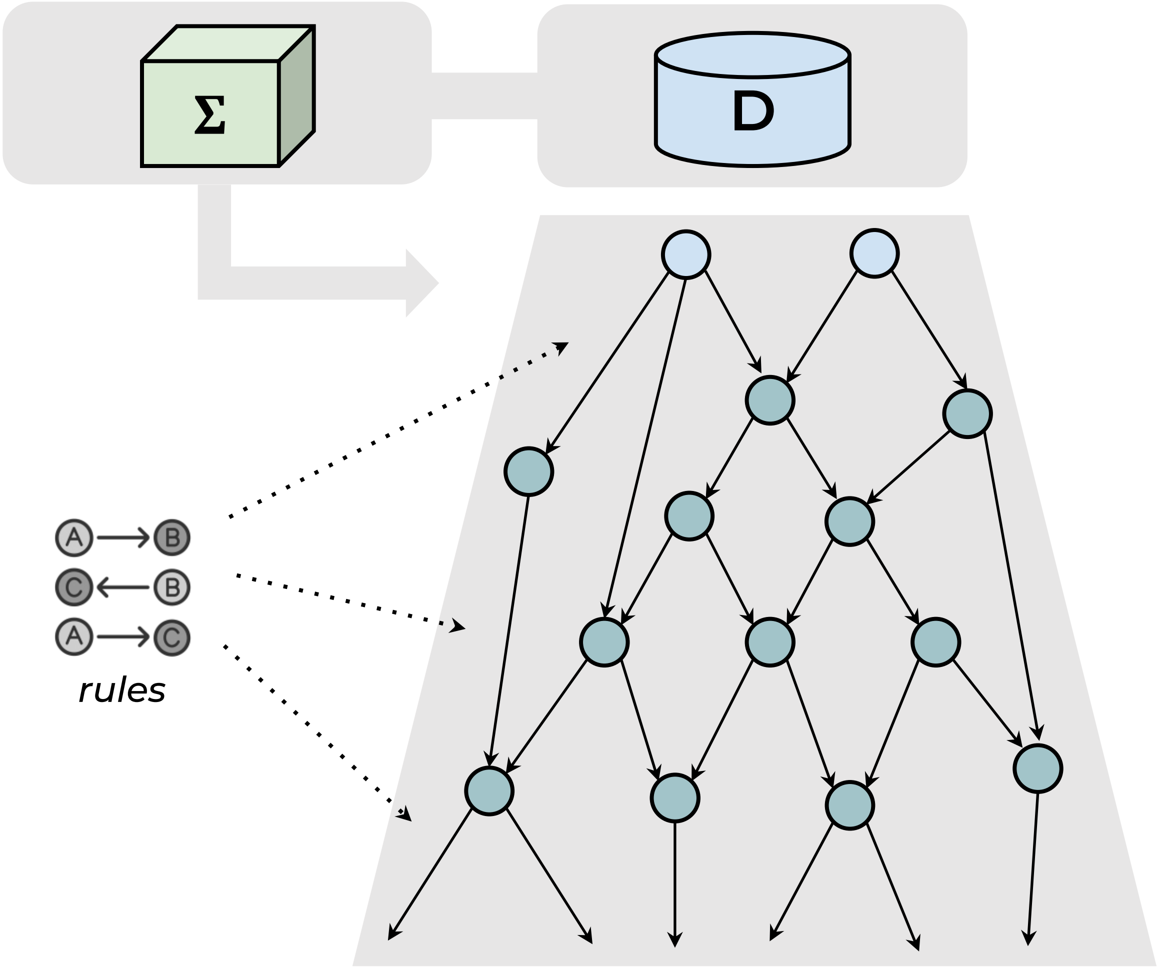
Shkapsky, A., Yang, M., & Zaniolo, C. (2015). Optimizing recursive queries with monotonic aggregates in deals. *ICDE*, pages 867-878.

Gottlob, G., & Pieris, A. (2015). Beyond SPARQL under OWL 2 QL entailment regime: Rules to the rescue. *IJCAI*.

- Semantics defined via the **chase** procedure, that takes in input a knowledge base **D** and a set **Σ** of rules and generates **new facts** by applying rules until **fixpoint**

- The chase is represented in the form of a **dependency graph** in which **nodes** are facts and **edges** represent rule applications to generate new facts





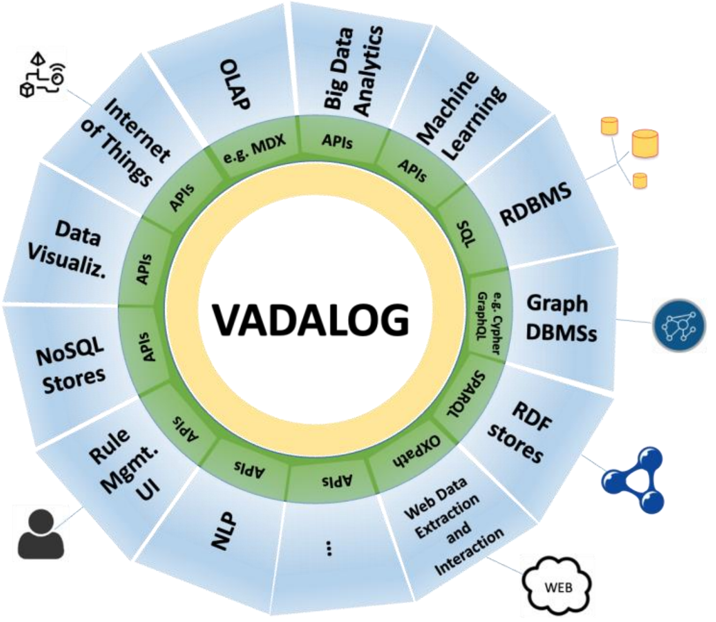
chase graph

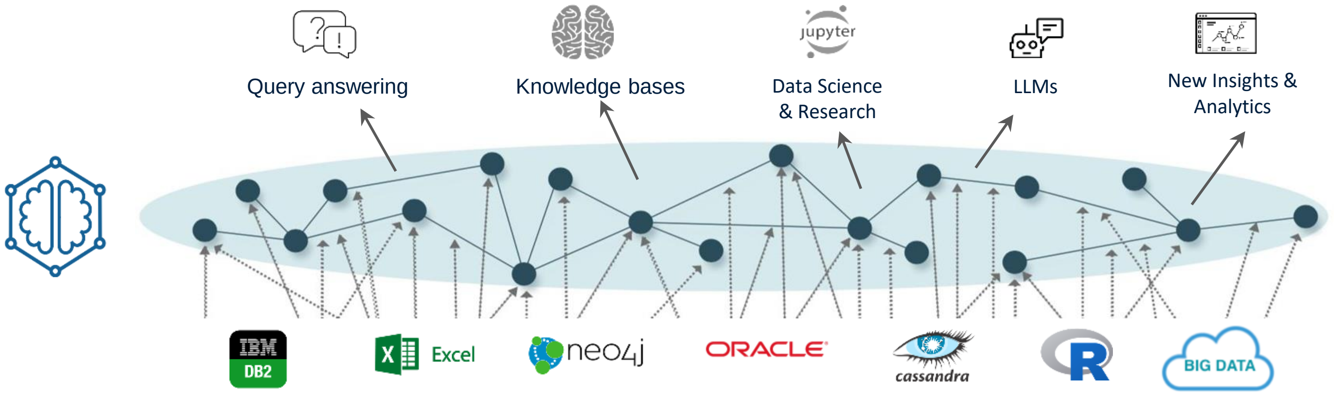
Maier, D., Mendelzon, A. O., & Sagiv, Y. (1979). Testing implications of data dependencies. TODS.
Shkapsky, A., Yang, M., & Zaniolo, C. (2015). "Optimizing recursive queries with monotonic aggregates in deals". IEEE.
Abiteboul, S., Hull, R., & Vianu, V. (1995). Foundations of databases (Vol. 8).

- The **Vadalog system** is a state-of-the-art reasoner that effectively addresses relevant and challenging tasks from **a plethora of domains** (from finance to genomics)

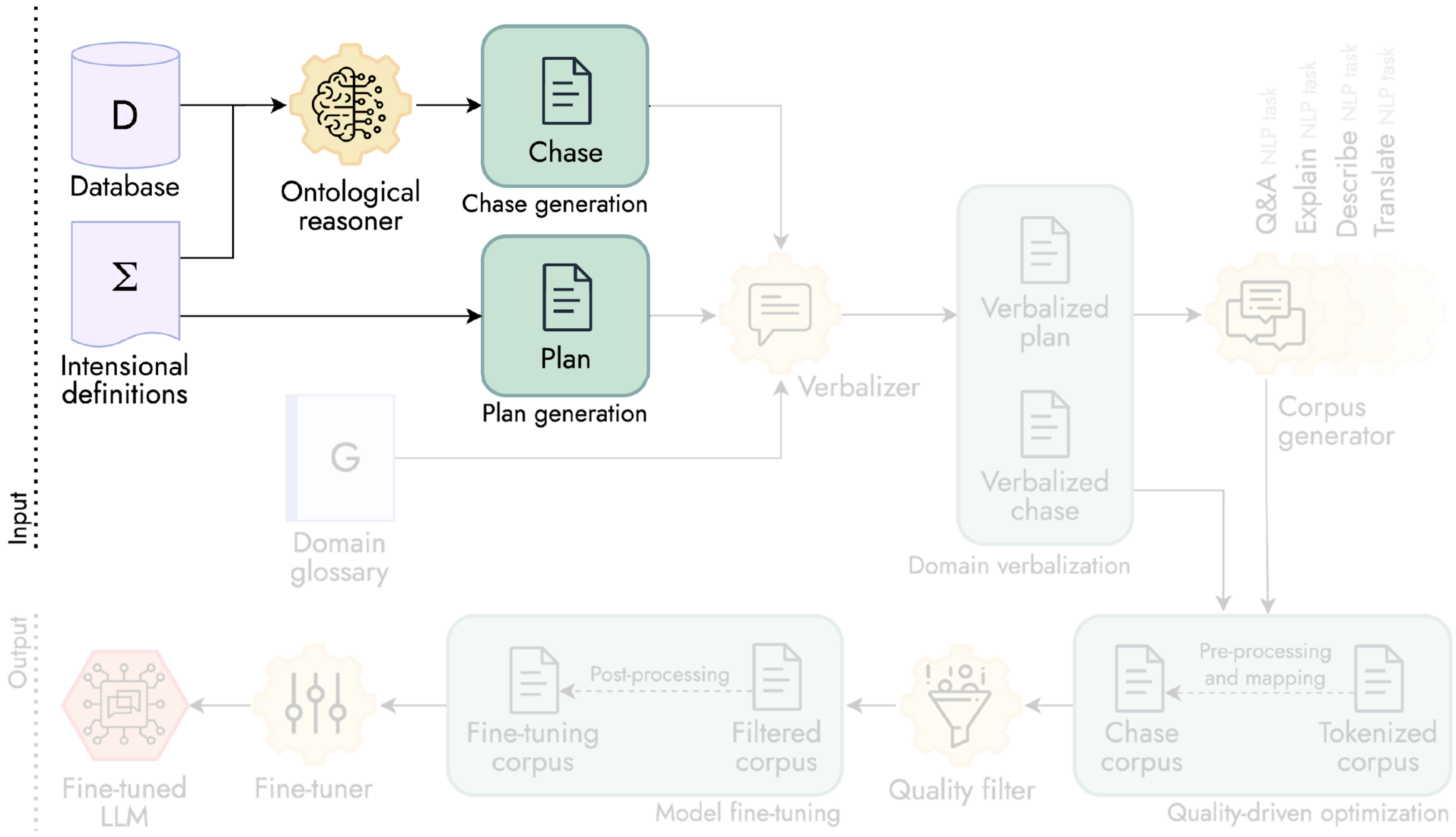
- It is developed by the **Joint KG Labs** as a joint effort of Banca d'Italia, TU Wien and University of Oxford

- It features Vadalog as **KRR language**, it supports multiple **data sources**, and it provides reasoning services via a **rich set of APIs**





Bellomarini, L., Benedetto, D., Gottlob, G., & Sallinger, E. (2020). Vadalog: A modern architecture for automated reasoning with large knowledge graphs. *Information Systems*, 101528.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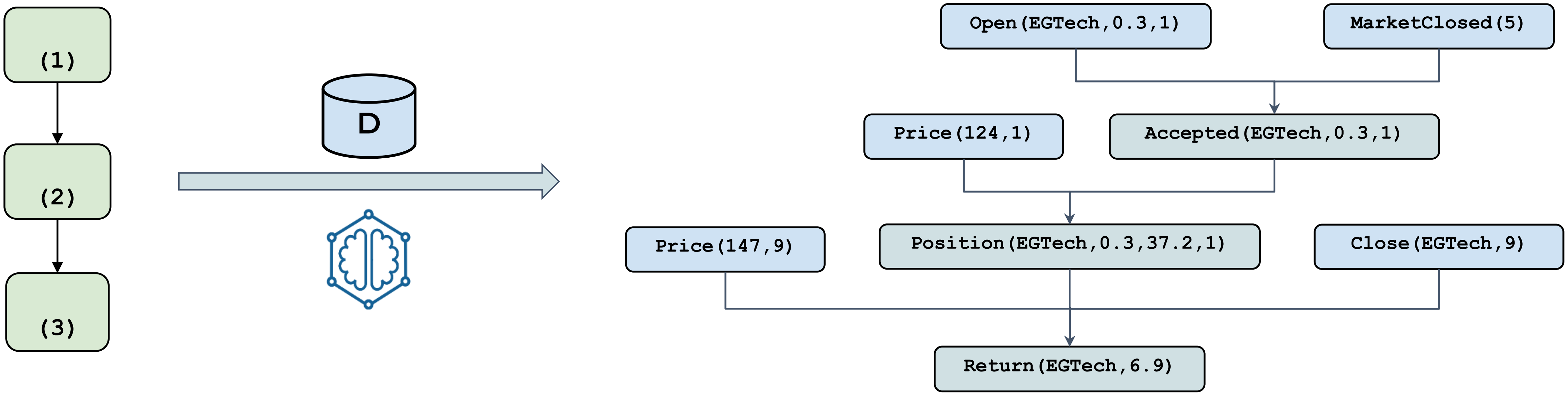
Simple trading activity with smart contract

$$\mathtt{Open}(x,y,t_1),\ \neg\ \mathtt{MarketClosed}(t_1) \rightarrow \mathtt{Accepted}(x,y,t_1) \quad (1)$$

$$\mathtt{Accepted}(x,y,t_1),\ \mathtt{Price}(p_1,t_1),\ k=y*p_1 \rightarrow \mathtt{Position}(x,y,k,t_1) \quad (2)$$

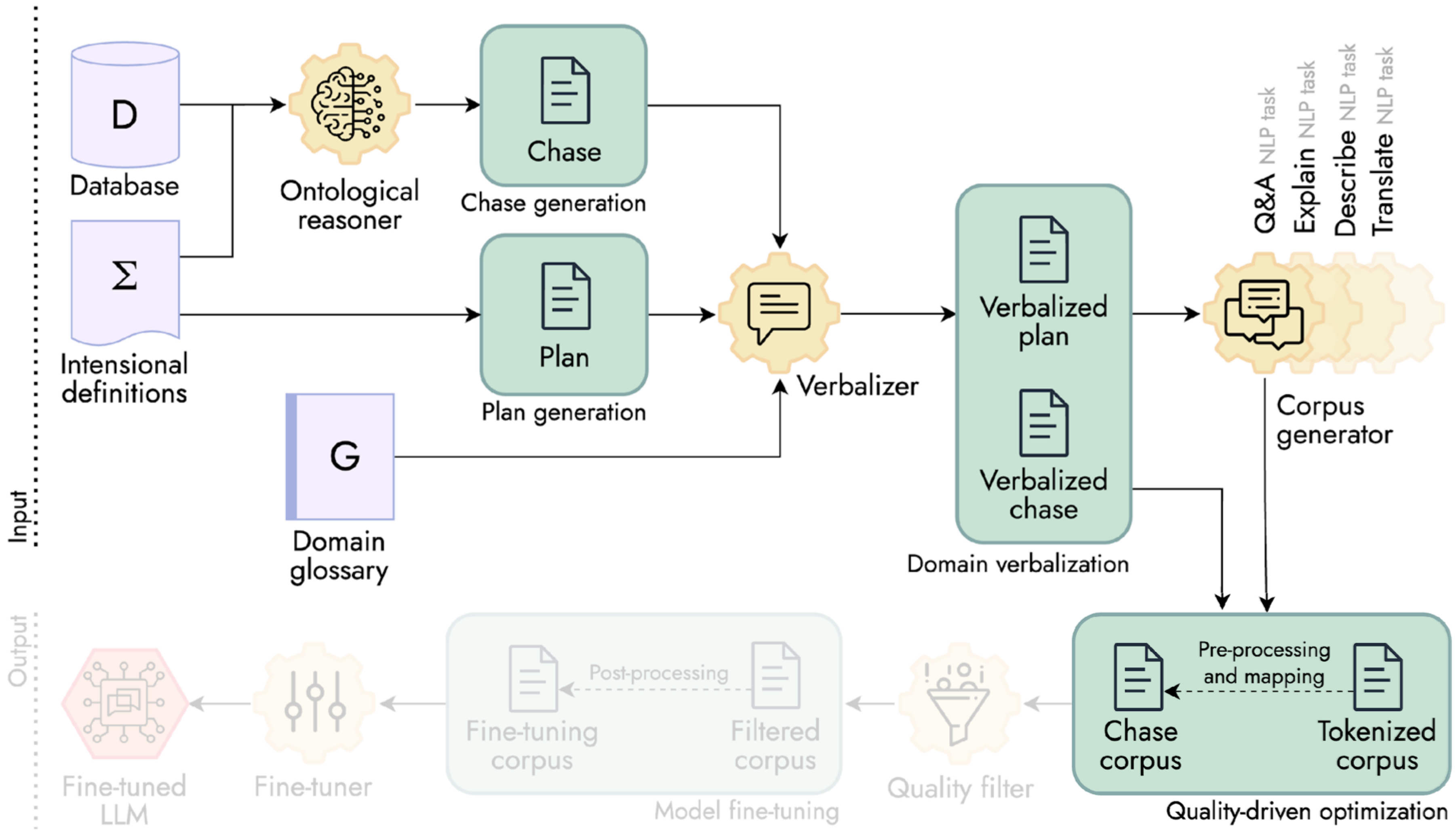
$$\mathtt{Close}(x,t_2),\ \mathtt{Price}(p_2,t_2),\ \mathtt{Position}(x,y,k,t_1),\ t_1>t_2,\ pl=y*p_2-k \rightarrow \mathtt{Return}(x,pl) \quad (3)$$

$$D = \{\mathtt{Open}(EGTech,0.3,1),\ \mathtt{Close}(EGTech,9),\ \mathtt{MarketClosed}(5),\ \mathtt{Price}(124,1),\ \mathtt{Price}(147,9)\}$$



logic plan

chase graph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Q&A NLP task

- the portions of the plan (i.e., templates) are **verbalized** into NL sentences via a **domain glossary**
- a pre-trained LLM generates **tokenized Q&A pairs** from the verbalized templates, sustaining **cost- and time-efficiency** as well as **protection of sensitive data**
- the arguments of the nodes in the chase graph are **mapped** to the Q&A corpus

$Open(x, y, t_1),$ $\neg MarketClosed(t_1)$ $\rightarrow Accepted(x, y, t_1)$ $\downarrow$ $Accepted(x, y, t_1),$ $Price(p_1, t_1), k = y * p_1$ $\rightarrow Position(x, y, k, t_1)$	Since the trader <x> at time <t_1> sends an order to open a position of size <y>, and it is not true that <t_1> is a time when the market is closed, then the order of size <y> by <x> is accepted at time <t_1>. Since the order of size <y> by <x> is accepted and the price is <p_1> at time <t_1>, then <x> holds a position of size <y> and notional <k> at time <t_1>.	Q1: When did <x> send an order to open a position with notional <k>? A1: The order to open that position was sent at time <t_1>	Q1: When did EGTech send an order to open a position with notional 37.2? A1: The order to open that position was sent at time 1
		…	…

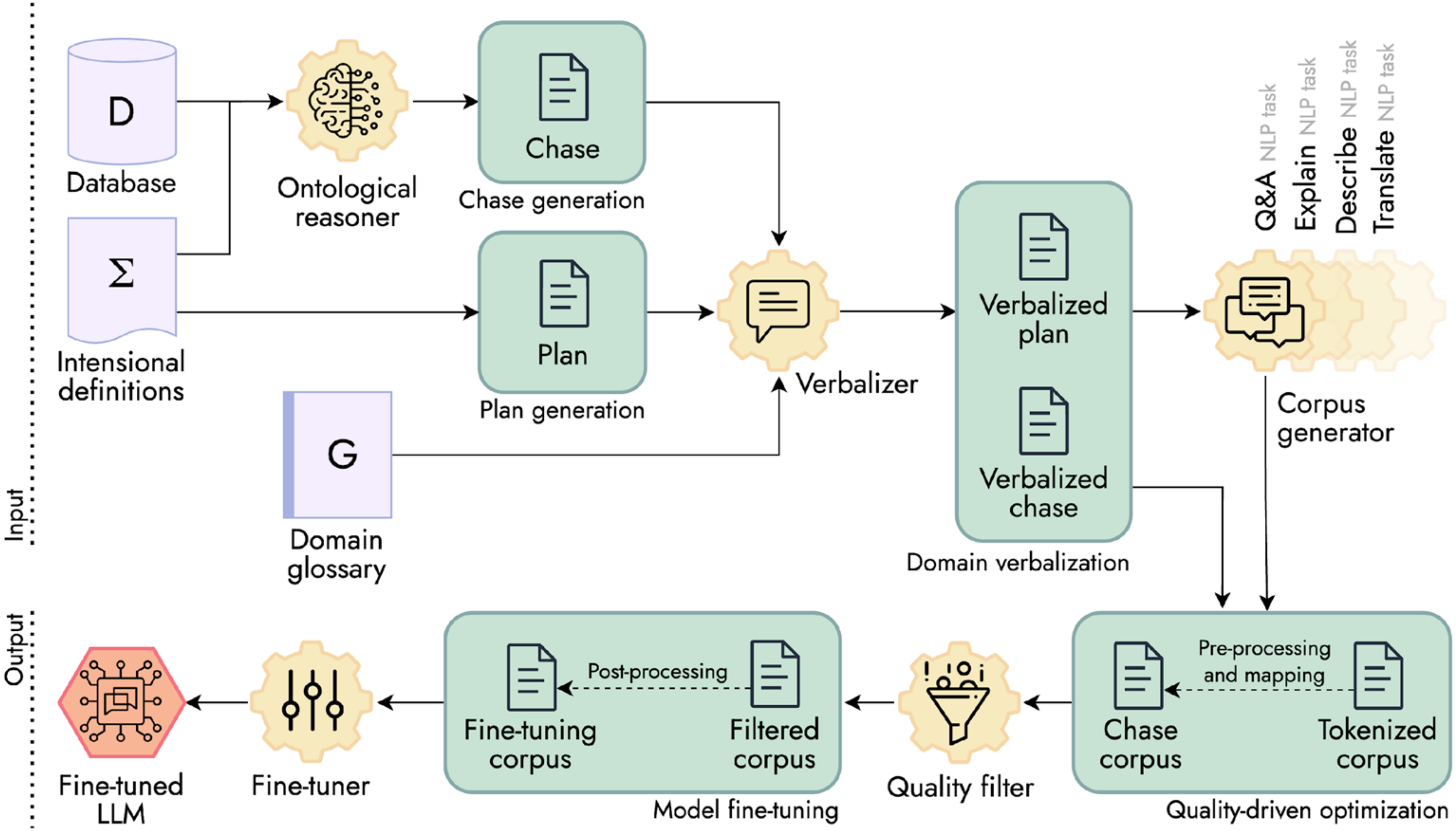
template verbalization

tokenized Q&A corpus generation

chase mapping to Q&A corpus

Domain glossary

Pre-trained LLM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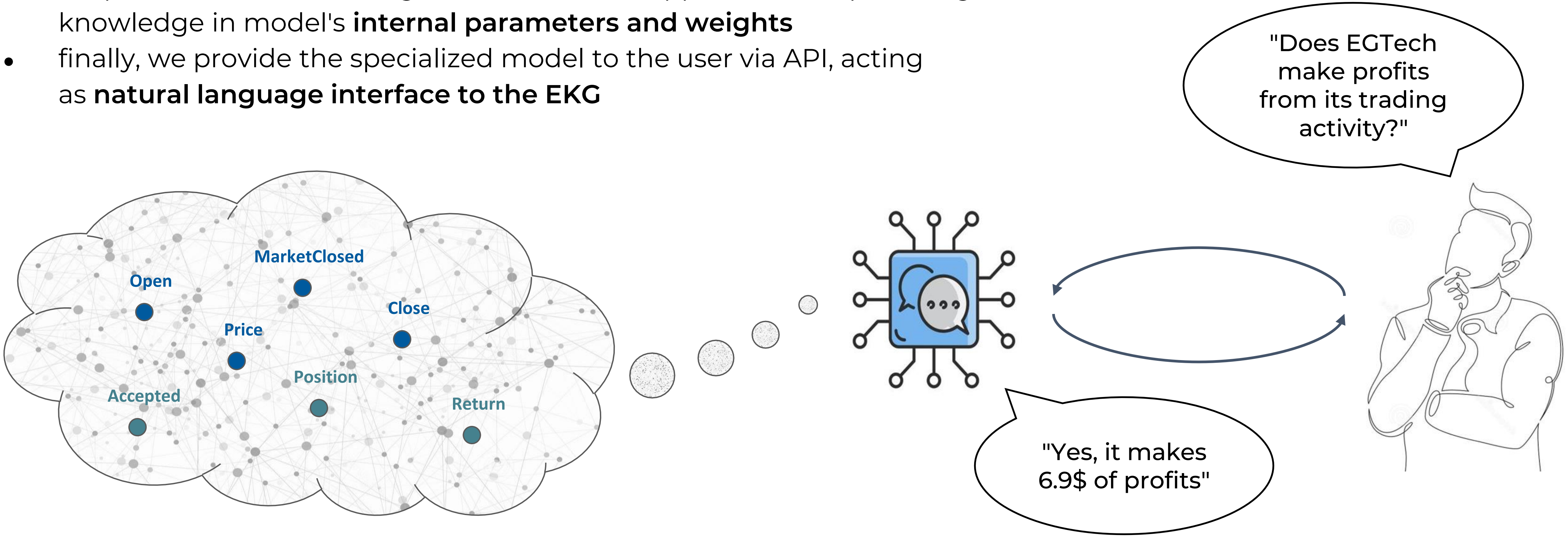
Q&A NLP task

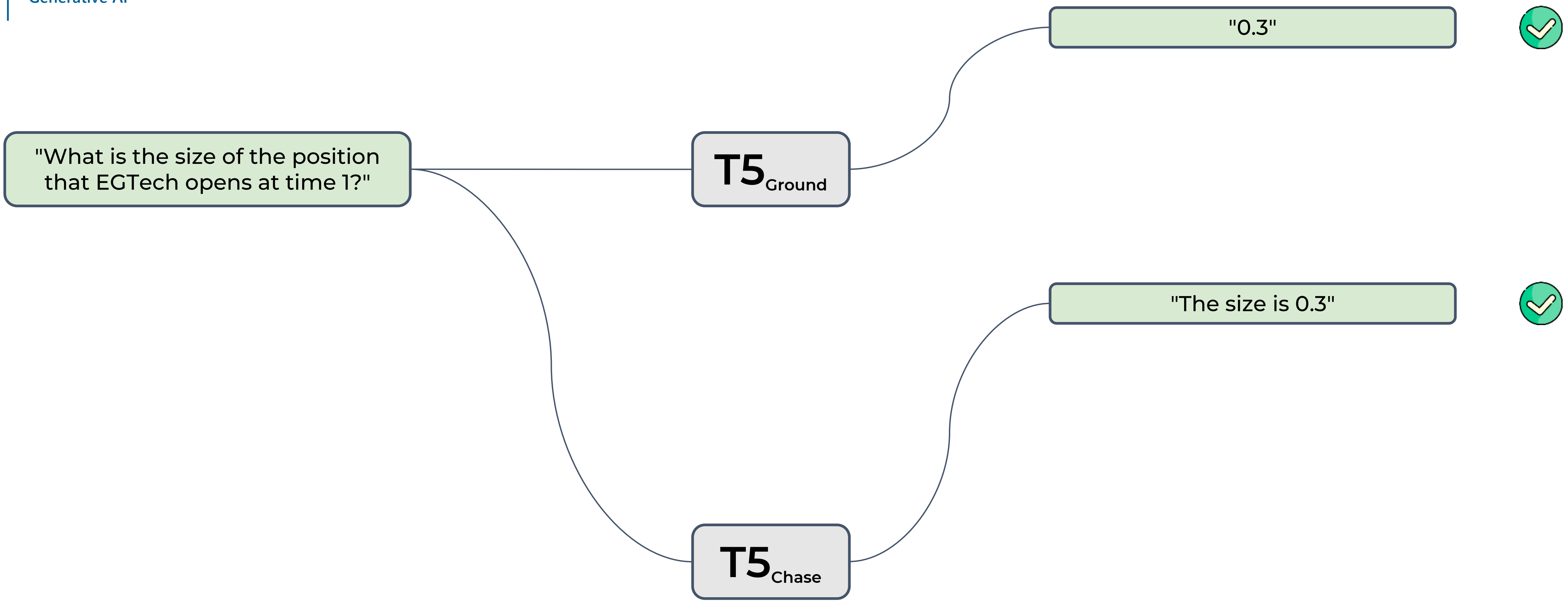
- we refine the corpus via **NLP paraphrasing** and other post-processing procedures to improve **quality** and **generalization** capabilities
- we perform the fine-tuning in a **closed-book** approach, encapsulating domain knowledge in model's **internal parameters and weights**
- finally, we provide the specialized model to the user via API, acting as **natural language interface to the EKG**



$\mathbf{T5}_{\mathrm{Ground}}$

$\mathbf{T5}_{\mathrm{Chase}}$

- $\mathbf{T5}_{\mathbf{Ground}}$ fine-tuned only on ground data
- $\mathbf{T5}_{\mathbf{Chase}}$ fine-tuned on chase data with our pipeline



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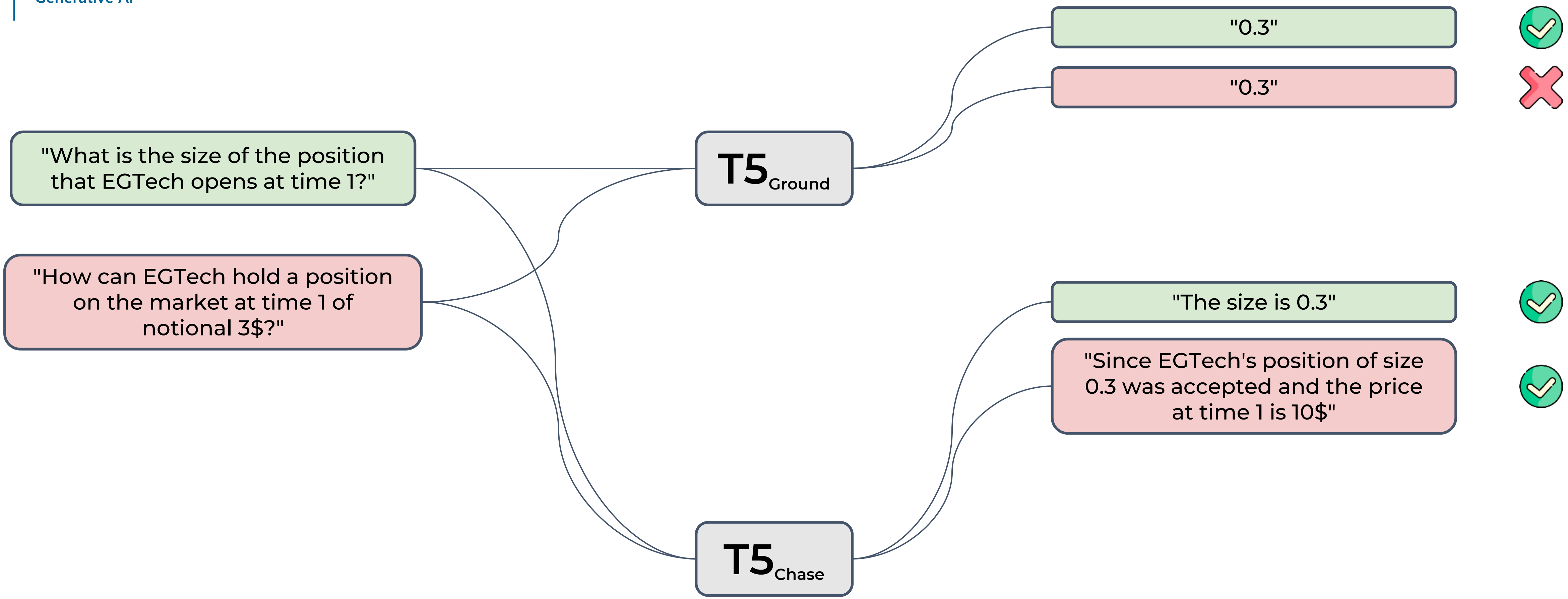

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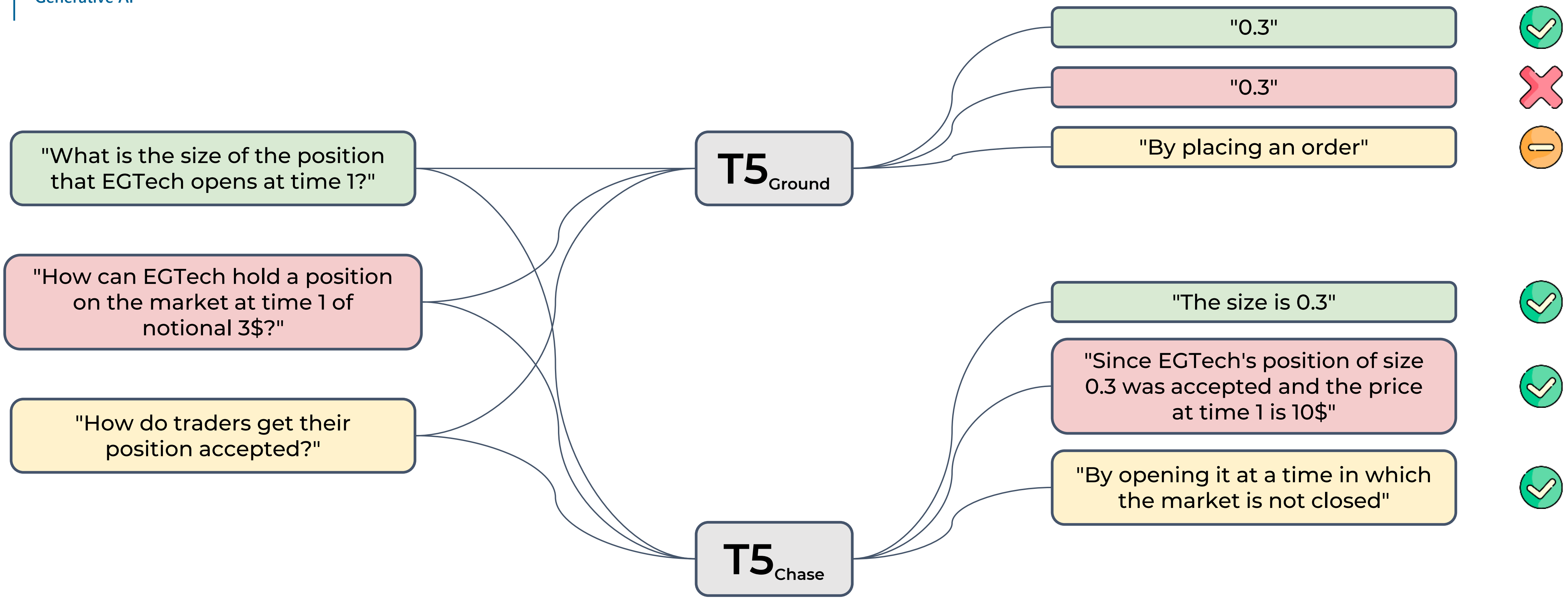

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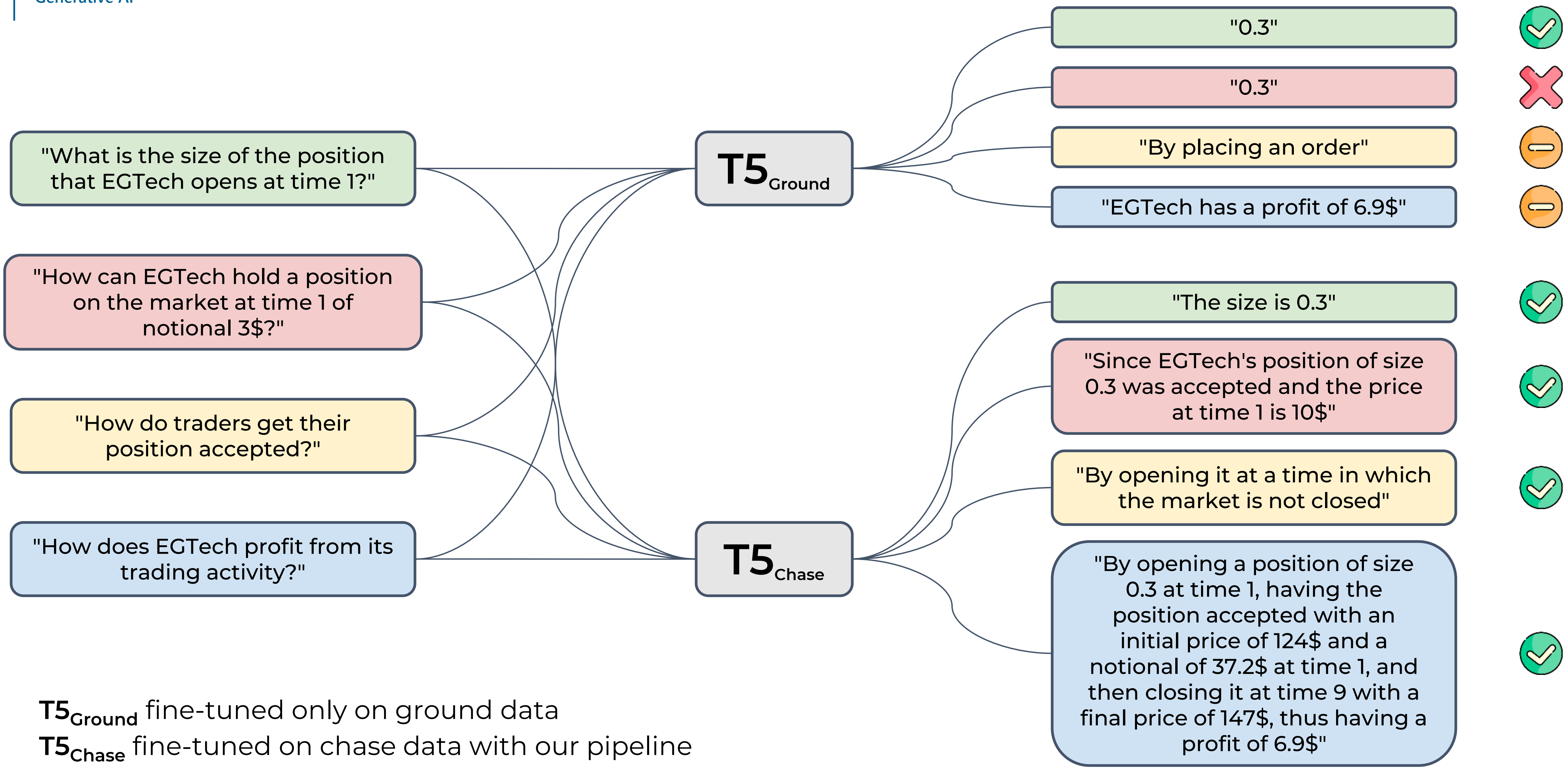

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- Pre-trained **LLMs cannot yet perform deduction**, being unable to generalize logical rules and use them to **infer and explain** new information

- LLM performance for **domain-specific tasks** can be visibly improved by producing a fine-tuning corpus as a **byproduct of ontological reasoning**

- The future is set to be **neurosymbolic**, based on a **synergistic interaction** between LLMs and ontological reasoning over **Enterprise Knowledge Graphs**

Yuan, Zhangdie, et al. (2023). Can Pretrained Language Models (Yet) Reason Deductively?. EACL
Dong, Xin Luna. (2023). Generations of Knowledge Graphs: The Crazy Ideas and the Business Impact. VLDB.

Knowledge Graphs × LLMs

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